

Learning Crop Models for Vision-Based Guidance of Agricultural Robots

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Abstract— This paper describes a vision-based method of guiding autonomous vehicles within crop rows in agricultural fields where the crop rows are challenging to detect or their appearance is not known a-priori. The location of the crop rows is estimated with an SVM regression algorithm using colour, texture and 3D structure descriptors from a forward facing stereo camera pair. Our system rapidly learns a model online with minimal user input, and then uses this model to track crop rows. Results demonstrate our method is able to learn and track a wide variety of crops with an RMS error of less than 3cm. We also present online control results demonstrating our system autonomously steering a robot for 3km.

I. INTRODUCTION

Automation of agricultural processes such as seeding, weeding, fertilising and harvesting has a key role to play in increasing farm productivity as well as meeting rising demands in food production. A key proficiency for most autonomous robots will be accurately guiding robots to drive with their wheels between the crop rows.

It is common for large agricultural vehicles to steer along pre-planned paths using precise Global Navigation Satellite System (GNSS) receivers. However, high-precision GNSS steering systems remain expensive, require real-time correction data as well as a precise map of crop row locations. This has encouraged a great deal of research into guiding agricultural vehicles and robots using local sensors such as cameras [1-12] which are low cost and can provide a rich source of information.

Current vision-based crop row tracking techniques typically segment green vegetation based on pixel colour, and then locate crop rows in the resulting binary image. This initial colour green vegetation segmentation step becomes problematic in some fields for example after harvest when only stubble (dead crop residue) remain, or when the crop has grown to form a near continuous green canopy.

In previous work [1] we presented a vision-only method that avoids segmentation entirely and instead tracks the offset of the parallel texture of the crop (see Figure 2). This method is able to guide a robot parallel to crop rows in a wide variety of fields and lighting conditions without prior knowledge of the crop appearance. This previous method however does not identify individual crop rows, which is important to prevent the wheels from damaging the crop. This is an issue we address in this work.

In this work we present a segmentation free method that uses a Support Vector Machine (SVM) regression algorithm to predict the offset of crop rows. We use an online semi-supervised training method with minimal human input to

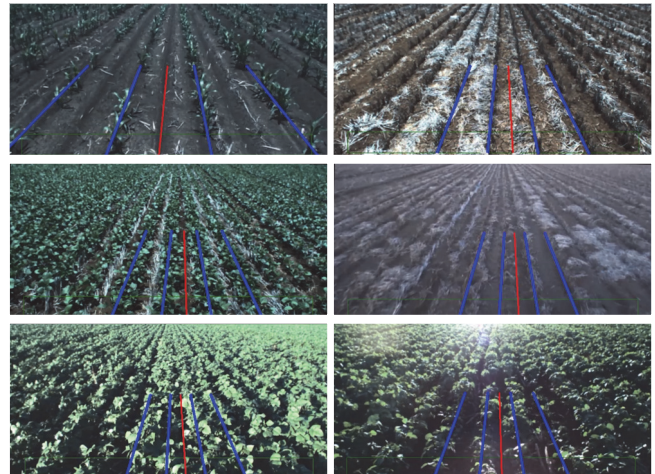


Figure 1: Six example images of fields demonstrating the large variability in appearance that occurs due to the crop type, stage of development and lighting conditions, making it difficult to find single algorithm to reliably identify crop rows. Our approach learns a model of crop rows for each field using colour, texture and 3D structure from stereo. Our system successfully tracks crop rows (marked with blue lines) and a path a vehicle can traverse without damaging the crop (marked with a red line) in each of these fields as without changing parameters.

rapidly learn a combined colour, texture and 3D structure model specific to that field. By learning a model for each field our system can handle crop rows with extremely varied appearance such as those shown in Figure 1. We demonstrate that this method can accurately track crop rows in a wide variety of fields even when crop rows are difficult to distinguish.

The paper is organised as follows. In Section II we describe related methods of visually navigating with respect to crop rows. Section III describes our method in detail, while Section IV and V presents our experimental setup and results from a variety of crops, before concluding remarks in Section VI.

II. RELATED WORK

Crop row tracking methods generally operate on the principal of either 3D structure or visual appearance.

Methods using 3D structure use either scanning lasers [2], [3], or stereo vision [4], [5] to identify the crop based on height. These methods are suitable for crops with an observable height difference between the crop and ground.

Many visual-appearance-based methods have been developed for machine guidance [6]–[8] and weed detection [9]–[11]. These methods perform an initial vegetation segmentation step, commonly by highlighting vegetation with a green index such as the Excess Green transform [12], then applying various adaptive segmentation methods such as Otsu thresholding [9] or k-means clustering [7]. The location of crop rows can then be located and tracked in this binary image

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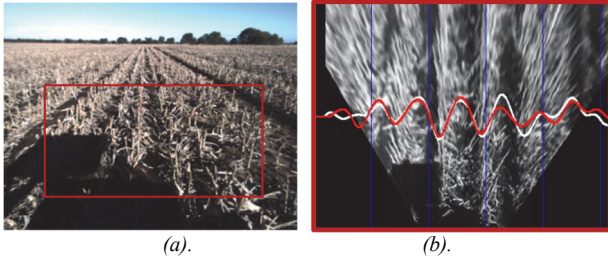


Figure 2: Brief overview of our previous texture based row-tracking method that forms a component of the system described in this paper. The red region of interest in (a) is transformed to an overhead view (b) before the direction of the rows is found and corrected by skewing the overhead image. Finally the relative offset is estimated by performing a zero-normalised cross-correlation of the average column intensity for the current and previous overhead images (red and white plots in b).

with a variety of techniques that are robust to outliers caused by weeds such as the Hough transform [6], [10], [11] or a Particle Filter [8].

The methods above generally use a plant identification strategy that is tailored to a specific set of crop types or stage of development based on a prior assumption, for example that the crop is green or tall. However, the appearance and structure of fields change dramatically from crop to crop and throughout the year, and it can be difficult to select a crop row tracking method suitable to all the situations that an agricultural robot is likely to encounter.

In previous work [1] we presented a vision-only crop row tracking algorithm that avoids segmenting crop from soil by tracking the parallel texture of the crop intensity. Figure 2 gives an overview of this method. We showed this method could reliably track the direction and relative offset of crops in varied and difficult conditions. While this method can guide a robot along crop rows, the method abstracts away the crop structure and so does not guide the vehicle to keep its wheels *between* the crop rows.

In this work we address this problem by using a machine learning algorithm to build a model of the crop rows from labelled examples, then use this model to predict the location (offset) of rows. The learning algorithm we use is a regression implementation of a Support Vector Machine (SVM) [13]. SVM's are a powerful and efficient supervised learning method that can fit complex non-linear models by mapping the input data into a higher dimensional feature space.

Our task is closely related to object tracking where the goal is to automatically track an object's location (bounding box) in a video stream. Discriminative models such as SVMs are often used for tracking in a technique called "Tracking by Detection" [14][15] which combines a classifier that detects the target with and a frame-to-frame tracker (such as [16]). We apply a similar concept in this work, where we combine an SVM regression algorithm to locate crop rows, with our previously described method [1] for frame-to-frame tracking.

III. ALGORITHM DESIGN

Given an image stream from a stereo camera pair mounted on a vehicle, and the location of one or more crop rows in the first frame, our goal is to locate the crop rows in all subsequent frames so that the vehicle can be steered with its wheels mid-

way between the crop rows. The procedure is explained in the following section. For an overview see Figure 3.

A. Frame-to-Frame Tracker

Our crop row tracking algorithm [1] processes images from the left camera of the stereo camera pair and outputs a reliable frame-to-frame relative crop row offset and heading for use by the rest of the system.

B. Overhead Image Patch

Input images are stabilised using IMU roll and pitch estimates. A small region of interest near the bottom of the image (the white rectangle in Figure 3.a) is then selected to capture a patch of ground in front of the robot. This gives a consistent and almost vertical viewpoint that is ideal for seeing through the crop canopy. We choose the height of this based on the typical vehicle speed and camera frame rate so as to provide full coverage of the ground while driving.

We use stereo matching to calculate the height above the ground of each pixel in the region of interest. Stereo matching is performed with the OpenCV implementation of Semi-Global Block Matching. This matching occurs quickly (< 50ms per frame) since we are processing only a small subset of the input image.

Finally, the RGB-height image is warped to a simulated overhead view using planar homography to remove perspective distortion and then skewed to correct for the crop direction. An example overhead image patch is shown in Figure 3.b.

C. Overhead Mosaic Image

The *overhead mosaic image* (Figure 3.c) combines overhead image patches (Figure 3.b) from a fixed number of previous frames to create a longer overhead image where crop rows are straight and parallel. Each overhead image is first translated to compensate for offset changes using the frame-to-frame tracker (from Section III.A), before being concatenated vertically. The height of this overhead mosaic image can be set to capture a large area of the field, allowing our system to cope with temporary variations in crop rows such as intermittent plants and bare patches. Typical dimensions for this overhead mosaic image are 600 pixels wide by 1200 pixels tall.

D. Feature Histogram Image

The overhead mosaic image is then converted into the *histogram image* shown in Figure 3.d which contains the image descriptors used later by the SVM regression.

We reduce the horizontal resolution of the overhead mosaic image by dividing it into blocks n columns wide. In this work we used $n = 10$ equating to a real world resolution of four centimetres.

We then calculate image descriptors for each block. The descriptors computed for each block are:

- colour histogram
- Local Binary Patterns (LBP)
- height histogram.

The descriptors calculated for each block (histogram bin counts) are normalised and unwrapped into a single column vector. These column vectors for each block are then

horizontally concatenated together to form a histogram image such as the one shown in Figure 3.d. That is, each column in the histogram image contains the descriptors for a corresponding n pixel wide block of pixels in the overhead mosaic image. This mapping can be observed in Figure 3, for example in the height histogram descriptors at the bottom of Figure 3.d the crop rows appear as periodic blue “blobs” that align with the crop rows in the overhead mosaic image in Figure 3.c.

Note that we chose descriptors that each compute a histogram as their output to keep the descriptor size fixed regardless of the input image height. For the colour histogram we use the transformed colour distribution defined as

$$\begin{pmatrix} R' \\ G' \\ B' \end{pmatrix} = \begin{pmatrix} \frac{R - \mu_R}{\sigma_R} \\ \frac{G - \mu_G}{\sigma_G} \\ \frac{B - \mu_B}{\sigma_B} \end{pmatrix}$$

where μ_c and μ_σ are the mean and standard deviation of the image under consideration. This colour transform is invariant to scaling and offsets in light intensity and colour [17]. We calculate a three dimensional colour histogram with 5 bins for each colour channel, for a total of $5 \times 5 \times 5 = 125$ colour histogram bins.

To capture texture information we use Linear Binary Patterns (LBP) texture descriptors. For each pixel we perform a binary comparison to 6 of its neighbours at a radius of 5 pixels, resulting in a 6 bit number. For each block we then create a 64 bin histogram to capture the $2^6 = 64$ unique binary texture values.

For the height histogram we quantise the height from stereo matching into 15 bins between ground height and the expected maximum crop height.

E. SVM Regression

SVM regression locates the crop rows. Given a window of values in the histogram image (the red region in Figure 3.d) the SVM regression predicts the distance (in pixels) to the nearest crop row. We use a sliding window approach to generate output predictions for each offset to produce the *predicted offset signal* in Figure 3.f. This signal is a unit-slope triangle wave that is zero when the sliding window is centred on a crop row, and at a maximum when the window is mid-way between crop rows. The width of the sliding window is usually chosen to be approximately equal to the crop row spacing.

While it is typical in tracking and object detection to use a binary *classifier* (such as an SVM classifier), in this work we have instead chosen to use a *regression* framework, which predicts a real valued number, rather than a binary class label. This is because the estimated output is a single real valued number (an offset) rather than a pair of numbers (x, and y position). We use regression to directly estimate our output, since we do not need to perform a multi-dimensional search (over x and y position) to maximise the output of a binary classifier, as is the case in classic 2D object detection.

The SVM regression implementation used is from OpenCV 2.4.9 We use an SVM type of “ ν -Support Vector Regressor” with Radial Basis Function (RBF) kernels, $\nu = 0.1$, $\gamma = 0.5$, and $C = 12.5$.

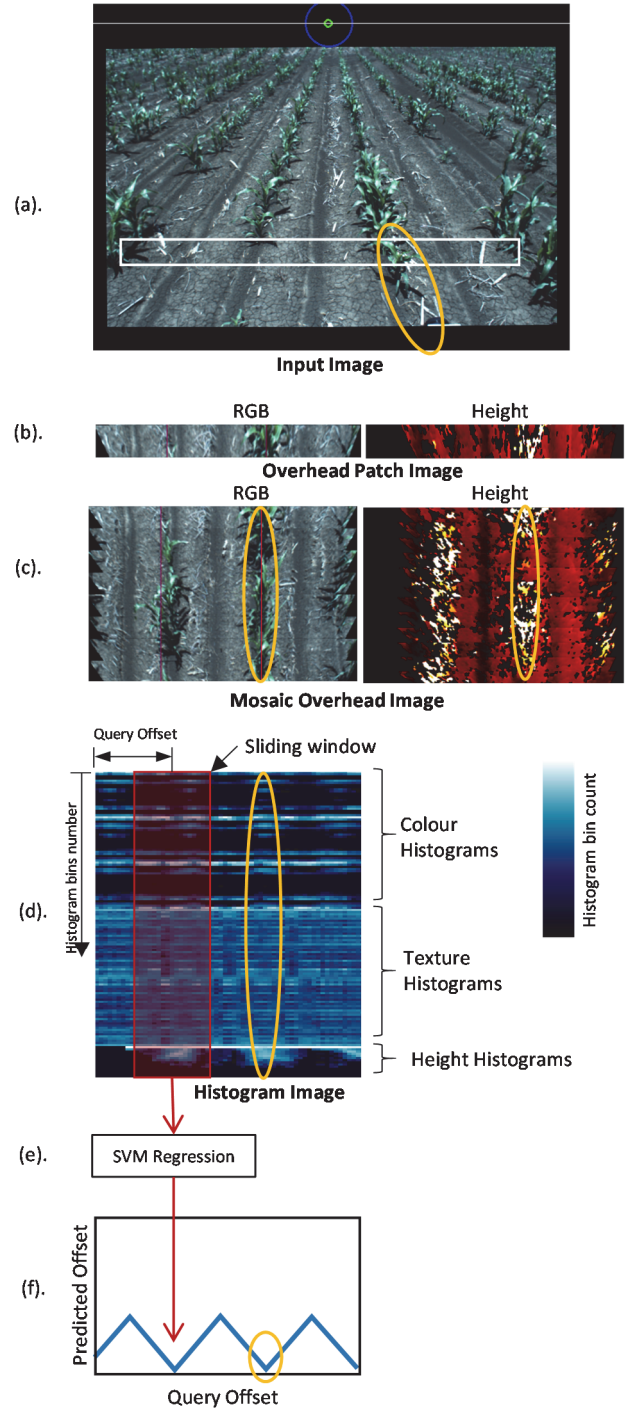


Figure 3: This figure shows how our algorithm predicts the crop row offsets. The orange ovals represent the same piece of crop in the various sub-figures. (a). Original input image with white region of interest rectangle. (b). Overhead view with RGB and height channels from the region of interest in the input image. (c). Overhead mosaic image created from successive overhead images. (d). Feature histogram image generated from overhead mosaic image. Each column corresponds to a group of columns in the overhead mosaic image. Rows correspond to separate histogram bins of the three descriptor types: colour, texture and height. Image intensity encodes the histogram bin count. (e). SVM trained to predict the offset to the nearest crop row. (f). Output from sliding window testing of the SVM. Crop rows are located at the minima of this plot. The location of crop rows and a steering signal is determined with further processing of these signals as described in Figure 4.

F. Output Filtering

To improve tracking performance, we also apply spatial and temporal post processing to the output of the SVM.

The predicted offset signal depicted in Figure 3.f is idealised and in practice we observe a noisier signal as shown in Figure 4.a. This makes locating the crop rows by at the local minima of this signal quite error prone. In order to more accurately locate the crop rows, we combine neighbouring offset predictions together by performing a template match of the predicted offset signal with the idealised template shown in Figure 4.b using a Sum of Squares Differences (SSD) similarity measure. The error output from this template matching produces a cleaner *SSD Error* signal shown in Figure 4.c. Note that the phase of this signal is opposite to the predicted offset, so crop rows are now located at the maxima of the SSD Error signal, and acceptable locations for our robot's wheels (mid-way between the crop rows) are located at the minima.

To further reduce noise, we also apply temporal filtering to the SSD Error signal with a simple complimentary filter, taking into account frame-to-frame offsets.

The filtered SSD Error is used to estimate the position of the crop rows and desired wheel location. We calculate a steering signal by locating the minimum value in a region around the known offset of the wheel nearest to our camera (determined via prior calibration).

G. Generating Training Data

The SVM is trained in a semi-supervised manner using labelled and unlabelled data. For a new field, two parameters are manually specified indicating the initial offset of two adjacent crop rows in the first frame. While driving in the field, these offsets are automatically transferred to subsequent frames using the frame-to-frame tracker (see Section III.A). This frame-to-frame tracker has very slow drift, usually less than 0.2% of the distance travelled, which allows the manually specified offsets from the first frame to be transferred to subsequent frames for a significant distance while still producing reasonable training examples. After training, the learned model is used to predict the crop row locations for the remainder of the run as well as for other rows within the same field.

We generate training examples for the SVM with the same sliding window approach used at test time so as to generate multiple training examples with varying offsets for each training image.

IV. CONFIDENCE MEASURE

We use a simple confidence measure to predict how well our SVM is able identify the crop rows. This confidence measure C is derived from the SSD Error signal E_{SSD} from Figure 4.c and is defined as

$$C = 1 - \frac{\min(E_{SSD})}{\max(E_{SSD})}.$$

This value ranges between 0 and 1 and we have anecdotally observed that tracking is generally successful when the confidence measure is greater than 0.5.

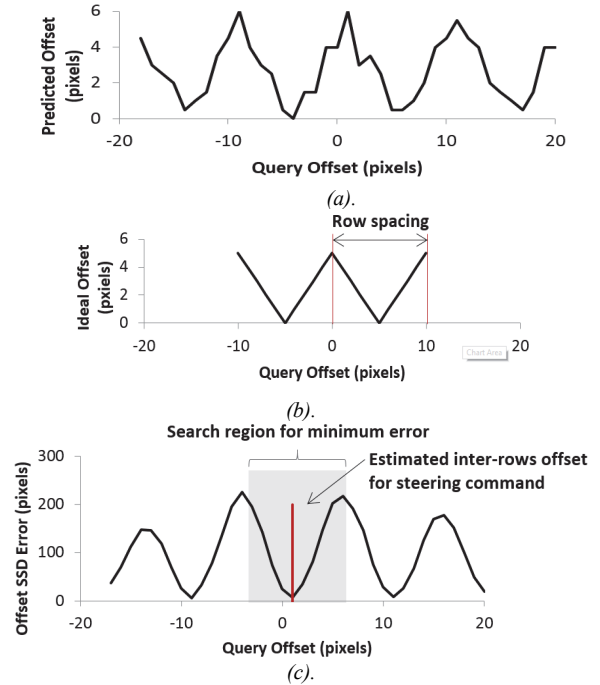


Figure 4: (a). Example of noisy predicted offsets. (b). Ideal offsets template. (c). Output of SSD template matching between predicted and ideal offset signals. Note this signal is much less noisy than the predicted offset in (a). Estimated inter-row positions are located at the maxima of the plot in (c).

V. EXPERIMENTAL SETUP

Experiments were performed on a broad-acre grain farm near Emerald, Australia, on a variety of cropping fields. Each experiment was performed on the robotic research platform shown in Figure 5. This vehicle is John Deere Gator electric vehicle converted for autonomous operation as described in our previous work [18]. The sensors used in these experiments were a Vectornav VN-100 Inertial Measurement Unit and a stereo pair of Point Grey GS3-U3-23S6C-C cameras with a field of view of 75 degrees, baseline of 700mm and capture rate of 10Hz. A high quality Novatel GNSS/INS system with RTK corrections via the SmartNetAus CORS network was used for recording ground truth position.

In the first experiment the vehicle was manually driven along crop rows in a range of fields and the data recorded for offline processing. In each case the initial position of the two crop rows either side of the left wheel was manually specified for the first frame. Our crop row detection method is run with training examples gathered automatically during the first 10 seconds of driving. To assess the accuracy of our system, the estimated inter-row position output by our system is compared against a manually labelled offset for each frame.

In the second experiment we perform an online control test where images were processed online at 10Hz on one of the robot's on-board PCs. We allow the vehicle to steer automatically along the crop rows at a velocity of 1.5m/s using a Proportional-Integral (PI) controller to convert the offset output by our method to a desired steering angle. This experiment was performed in the same wheat stubble field as test three in experiment one (depicted in Figure 6.c).

At the start of the experiment we placed the vehicle near the edge of the field with its wheels between the crop rows.

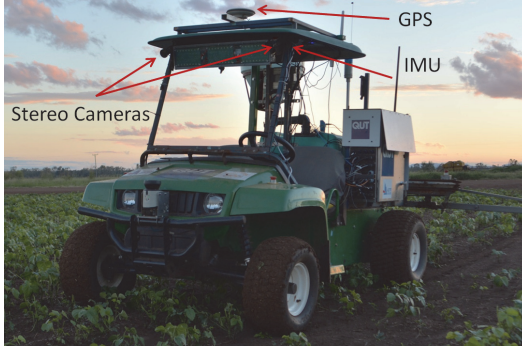


Figure 5: Robotic test platform used in experiments with sensors used labelled. High accuracy GPS/INS system is used for ground truthing only.

The location of the two crop rows adjacent to the left wheels was manually labelled in the first camera image. The vehicle then autonomously steered along the crop rows using the frame-to-frame offset while gathering training examples for 30 seconds. The vehicle then paused for approximately 20 seconds to learn a model, then subsequently used that model to steer along the remainder of the row.

At the ends of the field the vehicle was manually turned around and moved to a different crop row before re-engaging automatic control. The SVM was trained only on the first 30 seconds of data for the first run, with subsequent runs using the same model. This was to demonstrate that the model learned on one crop row is applicable to other rows despite different illumination due to travelling in the opposite direction.

In total we performed four runs in this experiment, accumulating approximately 2.8km of autonomous operation over 30 minutes. To evaluate the offset error we compared the path taken with the Novatel GNSS/INS system, since the crop rows were planted accurately in straight lines at a known angle with an RTK-GPS.

VI. RESULTS

A. Experiment 1 – Offline Processing

The section presents offline processed results from four fields: sorghum, mungbeans with wheat stubble, wheat

stubble, and mature mungbeans forming a near continuous canopy. Figure 6 shows an example image from each of these fields and a plot of the offset error for each test. Table 1 shows the RMS error for each of these tests. Note that each environment varies dramatically in appearance. In the first two fields the commonly used approach of green vegetation segmentation is clearly feasible, however in the last two examples green vegetation segmentation is unsuitable. Our method was successful in tracking in all cases with no changes in parameters.

TABLE I. OFFSET ERROR FOR OFFLINE EXPERIMENTS (M)

Test	RMS Offset Error (m)	95 th Percentile Offset Error(m)	Mean Confidence
Sorghum	0.026	0.047	0.87
Chickpeas w. stubble	0.024	0.052	0.73
Wheat Stubble	0.016	0.029	0.86
Chickpeas cont. canopy	0.031	0.059	0.73

These offset errors in the order of several centimetres. The 95th percentile errors range from about 5% of the crop row spacing for the first and third fields, to around 10% of the crop row spacing for second and fourth fields. It is thought that the additional error in the second and fourth fields (both mungbean crops) is due to the wider and less well defined crop rows. The fourth field (mature chickpeas) is challenging even for a human to identify the crop rows, however the results demonstrate that our system was able to select features to correctly identify the crop rows for the entire test.

B. Closed Loop Guidance

Figure 7 shows the offset error while the system is guiding the vehicle autonomously along the four different rows while Table 2 shows the errors for each run. The RMS error average is around 40mm which is approximately the same as the quantisation introduced when constructing the histogram features. It is thought that some of this error may also have come from the PI controller oscillating around the desired offset. Note that the offset error remains reasonably consistent between the four runs indicating that the model learned generalised well to other crop rows in the same field.

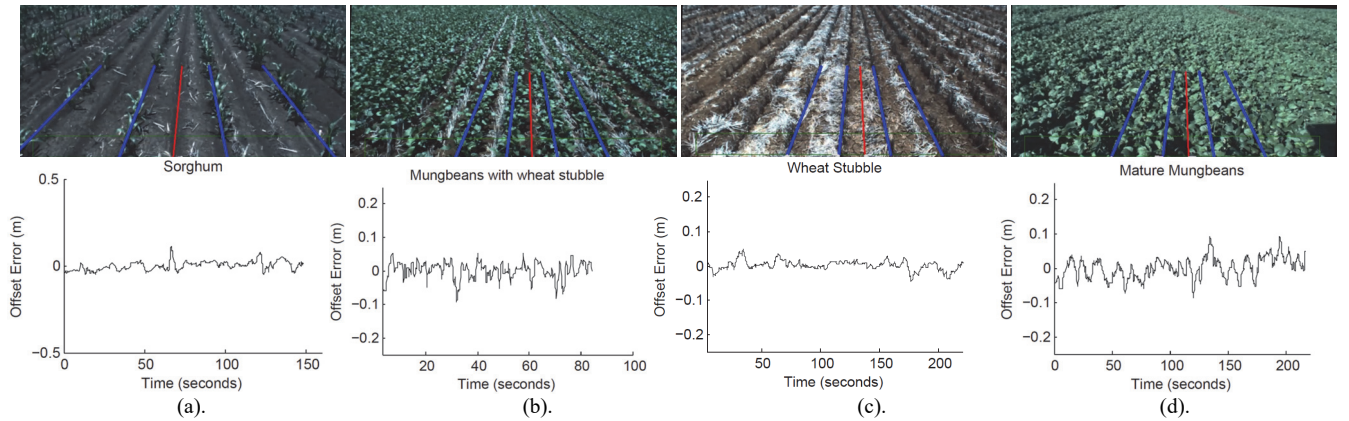


Figure 6: Results from offline experiment in fields with various crops. The first row shows an example frame from each field with the estimated crop row locations marked in blue, and the desired location for the robots wheel to drive, midway between crop rows, marked in red. The second row shows a plot of the offset error for each test. Offsets are measured in pixels, however have been converted into meters for clarity. The y axis in each graph is set to the distance between crop rows to give a sense of scale (1m for sorghum, and 0.5m for all other crops).

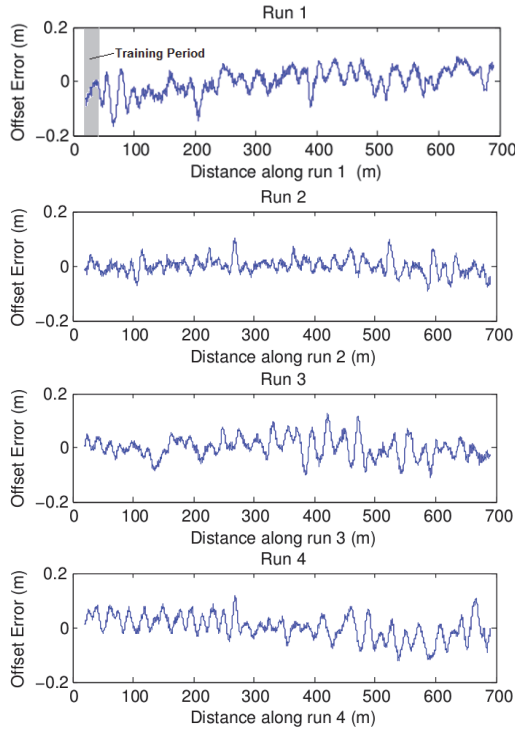


Figure 7: Offset error for each of the open loop experiment. The crop model was trained on data gathered during the first 30 seconds of the first run indicated by the highlighted region in (a). This model was subsequently used to the following three runs on different crop rows.

TABLE II. OFFSET ERROR CLOSED LOOP EXPERIMENTS (M).

Row Number	1	2	3	4	Total
RMS Error (mm)	0.045	0.028	0.039	0.044	0.040
95 th Percentile Error (mm)	0.065	0.047	0.065	0.068	0.063

VII. DISCUSSION AND CONCLUSION

This paper demonstrates a method of visually identifying the offset of crop rows in agricultural fields using a learning algorithm to rapidly learn a model applicable to a specific field in a quick and practical way with minimal human input. Results demonstrate it is applicable to a wide variety of fields and crops, and is able to perform online guidance of a robot.

Our method currently uses a one-off batch mode approach to training. In future work we would like to investigate methods of extending this method to performing continuous online learning to allow adaption to changing lighting and field conditions. One possible scheme we would like to investigate to allow online adaption is to temporarily stop the robot when our confidence measure falls below a threshold and to signal a remote operator to indicate the offset of one or more crop rows in a single image. This small amount of human input could then automatically generate many additional training examples that could be added to the original training set to retrain the model to also generalise to this new field appearance. While this scheme requires some occasional human input, it is hoped that over time the robot will gather enough training examples for common lighting conditions and crop appearances that the crop model could be general enough to no longer require human input.

We would also like to investigate more powerful image descriptors and learning algorithms to enable better discriminative power, and would like to combine the output of our system with a low-cost GPS within a particle filter framework to create a low-cost and accurate navigation system for agricultural robots.

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